AS 1720.2—2006
(Incorporating Amendment No. 1)

AS 1720.2—2006

Australian Standard™

Timber structures

Part 2: Timber properties

STANDARDS
Australia

This Australian Standard was prepared by Committee TM-001, Timber Structures. It was approved on behalf of the Council of Standards Australia on 8 November 2005.
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Association of Consulting Engineers Australia
Australian Building Codes Board
Australian Timber Importers' Federation
Australian Wood Panels Association
Building Research Association of New Zealand
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PREFACE

This Standard was prepared by the Joint Standards Australia/Standards New Zealand Committee TM/1, Timber Structures, to supersede AS 1720.2—1990. After consultation with stakeholders in both countries, Standards Australia and Standards New Zealand decided to develop this Standard as an Australian Standard rather than an Australian/New Zealand Standard.

This Standard incorporates Amendment No. 1 (April 2006). The changes required by the Amendment are indicated in the text by a marginal bar and amendment number against the clause, note, table, figure or part thereof affected.

The objective of this Standard is to provide a list of structural timbers and their properties. Information on the availability and source of particular species is normally available from local industry associations.

The main changes in this edition are as follows:

(a) Inclusion of structural classification, as the new Appendix A, Tables A1 and A2, previously in AS 1720.1—1988.

(b) Amendment of timber natural durability ratings to align with those given in AS 5604.

This Standard is the second part in a series of timber structures Standards, which is comprised of the following:

AS
1720 Timber structures
1720.1 Part 1: Design methods
1720.2 Part 2: Timber properties (this Standard)
1720.4 Part 4: Fire-resistance of structural timber members

The term 'informative' has been used in this Standard to define the application of the appendix to which it applies. An 'informative' appendix is only for information and guidance.

CONTENTS

Page

1 SCOPE AND GENERAL 4
2 REFERENCED DOCUMENTS 4
3 TABULATION OF SPECIES AND PROPERTIES 4
4 EXPLANATION OF TABLE 1 8
5 REFERENCES 13

APPENDIX A F-GRADES FOR VISUALLY GRADED TIMBER 14

STANDARDS AUSTRALIA

Australian Standard
Timber structures

Part 2: Timber properties

1 SCOPE AND GENERAL

This Standard sets out a table of species and their general properties, which can be used for the design of timber structures.

NOTES:

1 It is emphasized that structural timber exhibits considerable natural variability and, consequently, the information given herein might be inadequate when precise and specific data are required.

2 The inclusion of a species in this Part does not necessarily indicate that it is readily available.

2 REFERENCED DOCUMENTS

The following documents are referred to in this Standard:

AS

A1 1649 Timber—Methods of test for mechanical fasteners and connectors—Basic working loads and characteristic strengths

1720 Timber structures

1720.1 Part 1: Design methods

2082 Timber—Hardwood—Visually stress-graded for structural purposes

2209 Timber—Poles for overhead lines

2858 Timber—Softwood—Visually stress-graded for structural purposes

2878 Timber—Classification into strength groups

5604 Timber—Natural durability ratings

AS/NZS

1148 Timber—Nomenclature—Australian, New Zealand and imported species

4063 Timber—Stress-graded—In-grade strength and stiffness evaluation

ASTM

D143 Standard Test Methods for Small Clear Specimens of Timber

BS

373 Methods of testing small clear specimens of timber

3 TABULATION OF SPECIES AND PROPERTIES

Some species used for structural purposes are listed in Table 1 together with their classification and general properties.

NOTE: See Appendix A for information regarding assignment of F-grades to visually graded timber.

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TABLE 1

STANDARD TRADE NAMES AND GENERAL PROPERTIES OF SOME COMMON SPECIES

1	2	3	4	5	6		7		8		9		10		11		12	13	14	15		16	17	18
Index No.	Standard trade name	Origin	Species brand	Classification	Strength group		Joint group		Design density† (kg/m³)		Janka hardness, side grain (kN)		Tangential cleavage (N/mm)		Toughness (N·m)		Tangential % shrinkage	Tendency to split for unseasoned timber	% Unit tangential movement	Natural durability class§		Sapwood lyctus susceptible	Termite resistant	Marine borer durability
					G	D	G	D	G	D	G	D	G	D	G	D				In-ground contact	Outside above ground			
8	alder, rose	Q	RA	H	S6	SD6	J4	JD4	1050	550	—	—	—	—	—	—	5.2‡	—	0.23	4	—	NS	NR	4
22	ash, alpine	N, T, V	AA	H	S4	SD4	J3	JD3	1050	650	4.0	5.0	60.4	70.2	17.1	18.9	8.5‡	high	0.35	4	3	S(N, T), NS(V)	NR	4
30	ash, mountain	T, V	MA	H	S4	SD3	J3	JD3	1050	650	3.4	4.9	49.9	72.6	17.0	20.0	13.3‡	high	0.36	4	3	NS	NR	4
37	ash, silvertop	N, T, V	ST	H	S3	SD3	J2	JD2	1100	850	6.7	9.7	79.3	101.0	22.6	22.6	10.6‡	high	0.36	3	2	NS	NR	4
84	blackbutt	N, Q	BB	H	S2	SD2	J2	JD2	1150	900	6.4	8.9	70.0	77.0	21.1	22.0	7.3‡	medium	0.37	2	1	NS	R	3
86	blackbutt, New England	N, Q	NA	H	S3	SD3	J2	JD2	1150	850	6.6	9.2	84.5	86.4	16.9	20.7	11.4‡	high	0.36	2	2	S	R	2
87	blackbutt, W.A.	W	BA	H	S4	SD5	J2	JD2	1100	850	5.5	6.9	—	—	14.2	11.5	—	—	—	2	1	S	R	3
121	box, brush	N, Q, Y	BH	H	S3	SD3	J2	JD2	1150	900	7.8	9.1	83.3	110.4	20.7	16.2	9.7‡	high	0.38	3	3	NS	R	2
127	box, grey, coast	N, V	CB	H	S1	SD1	J1	JD1	1200	1100	10.3	12.8	95.6	90.6	28.0	31.1	8.2	medium	0.42	1	1	S	R	3
150	box, yellow	N, Q, V	YB	H	S3	SD4	J1	JD1	1200	1050	10.5	13.0	92.6	89.4	21.5	16.0	6.2	low	0.39	1	1	NS	R	—
165	brownbarrel	N, V	BL	H	S4	SD4	J3	JD3	1100	750	5.0	5.5	63.3	86.1	18.9	18.1	11.8‡	high	0.34	4	3	S	NR	4
178	candlebark	N, S, T, V	CD	H	S5	SD5	J3	JD3	1100	700	5.0	5.9	58.4	63.7	16.2	11.7	12.2‡	high	0.34	4	3	S	NR	4
114	fir, Douglas	Nth Amer.	DF	S	S5	SD5	J4	JD4	700	550	2.1	3.0	38.0	39.0	10.8	12.0	4.0	low	—	4	4	NS	NR	4
		NZ(P)			S6	SD6	J5	JD5	800	420	2.2	3.6	43.8	42.0	—	—	4.9	Low	—					—
253	gum, blue, southern	N, T, V	SB	H	S3	SD2	J2	JD2	1150	950	7.3	11.4	83.6	111.1	23.7	26.5	7.7‡	medium	0.40	3	2	S	NR	4
254	gum, blue, Sydney	N, Q	SY	H	S3	SD3	J2	JD2	1100	850	5.8	8.1	71.8	84.0	17.8	18.4	9.5	high	0.35	3	2	S	NR	3
266	gum, grey	N,Q	GG	H	S1	SD2	J1	JD1	1150	1050	—	—	—	—	—	—	—	—	—	1	1	NS	R	2
267	gum, grey, mountain	N,V	MT	H	S3	SD2	J2	JD2	1100	850	6.5	9.9	81.9	101.7	20.4	20.0	11.9‡	high	0.39	3	2	S	NR	4
269	gum, manna	N,Q,S,T,V	MN	H	S4	SD4	J3	JD2	1100	750	5.1	5.8	65.8	73.5	17.8	16.0	12.0‡	high	0.34	4	3	S	NR	4
272	gum, mountain	N,T,V,Q	MO	H	S4	SD5	J3	JD3	1100	700	5.2	5.7	65.3	72.3	17.9	17.4	11.5‡	high	0.35	4	3	S	NR	4
279	gum, red, forest	N, Q, V, G	FR	H	S3	SD4	J1	JD1	1150	1000	8.6	11.3	76.3	98.5	19.4	17.6	8.6	high	0.34	1	1	NS	R	2
281	gum, red, river	N, Q, S, V, W, Y	RR	H	S5	SD5	J2	JD2	1100	900	6.8	9.6	81.0	94.8	15.6	10.6	8.9‡	high	0.31	2	1	S	R	2
284	gum, rose	N, Q	RO	H	S3	SD4	J2	JD2	1100	750	5.5	7.3	73.3	88.7	22.4	18.5	7.5	medium	0.30	3	2	NS	NR	4
289	gum, shining	N, V	SH	H	S4	SD4	J3	JD3	1100	650	4.8	5.8	70.9	57.8	17.4	15.8	9.4‡	high	0.33	4	3	S	NR	4
293	gum, spotted	N, Q, V	SG	H	S2	SD2	J1	JD1	1200	1000	8.8	10.1	78.8	86.1	25.3	27.5	6.1	low	0.38	2	1	S	R	4

(*continued*)

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TABLE 1 (*continued*)

1	2	3	4	5	6		7		8		9		10		11		12	13	14	15		16	17	18
Index No.	Standard trade name	Origin	Species brand	Classification	Strength group		Joint group		Design density† (kg/m³)		Janka hardness, side grain (kN)		Tangential cleavage (N/mm)		Toughness (N·m)		Tangential % shrinkage	Tendency to split for unseasoned timber	% Unit tangential movement	Natural durability class§		Sapwood lyctus susceptible	Termite resistant	Marine borer durability
					G	D	G	D	G	D	G	D	G	D	G	D				In-ground contact	Outside above ground			
310	hardwood, Johnstone River	Q	JR	H	S2	SD3	J1	JD1	1150	1000	—	—	—	—	—	—	6.4	—	0.39	3	2	S	NR	4
139*	hemlock	Nth Amer	WH	S	S6	SD6	J4	JD4	750	500	2.1	2.7	35.4	27.5	9.2	9.2	5.0	medium	—	4	4	NS	NR	4
322	ironbark, grey	N, Q	GI	H	S1	SD1	J1	JD1	1250	1100	10.8	16.3	87.3	78.4	30.0	33.0	7.5	medium	0.39	1	1	NS	R	3
325	ironbark, red	N, Q, V	RI	H	S2	SD3	J1	JD1	1200	1050	10.7	11.9	93.4	74.6	26.0	23.0	6.3	low	0.37	1	1	S	R	2
326	ironbark, red, broad-leaved	N, Q	BI	H	S1	SD1	J1	JD1	1200	1100	12.0	14.0	—	—	—	—	—	—	—	1	1	NS	R	—
327	ironbark, red, narrow-leaved	N, Q	NI	H	S2	SD3	J1	JD1	1250	1050	12.0	14.0	—	—	—	—	—	—	—	1	1	NS	R	—
341	jarrah	W	J	H	S4	SD4	J2	JD2	1100	800	5.7	8.5	67.4	81.2	14.2	13.2	7.4	medium	0.30	2	2	S	R	3
154*	kapur	S. E. Asia	KR	H	S3	SD4	J2	JD2	1100	750	4.0	5.4	55.0	62.0	13.8	15.4	6.0	medium	—	3	2	NS	NR	4
344	karri	W	K	H	S3	SD2	J2	JD2	1150	900	6.0	9.0	80.5	74.9	22.6	23.0	9.9	high	0.40	3	2	NS	NR	4
169*	keruing	S. E. Asia	KG	H	S3	SD3	J2	JD2	—	780	3.2	4.6	44.0	62.0	26.7	32.5	9.5	high	—	3	3	S	NR	4
173*	kwila (merbau)	S. E. Asia	KW	H	S2	SD3	J2	JD2	1150	850	7.6	8.8	78.0	87.0	21.6	19.4	2.5	low	—	3	1	S	R	3
384	mahogany, red	N, Q	RM	H	S2	SD3	J1	JD1	1200	950	9.1	12.0	67.0	75.8	17.4	19.1	6.3	medium	0.34	2	1	S	R	2
387	mahogany, southern	N, V	SM	H	S2	SD3	J2	JD2	1100	900	7.0	9.4	75.2	84.5	18.1	19.9	9.8‡	high	0.37	3	2	NS	R	—
391	mahogany, white	N, Q	WM	H	S2	SD3	J1	JD2	1150	950	8.9	10.1	66.0	—	19.7	19.2	5.4	low	—	1	1	NS	R	2
432	marri	W	ME	H	S3	SD3	J2	JD2	1100	850	6.6	7.1	69.8	75.8	21.4	23.2	6.6	medium	0.34	3	3	S	NR	4
221*	meranti, dark red	S. E. Asia	DM	H	S5	SD6	J4	JD3	1100	650	3.1	3.5	47.0	—	—	—	5.0	low	—	4	3	S	NR	4
434	messmate	N,Q,S,T,V	MS	H	S3	SD3	J3	JD3	1100	750	5.3	7.2	64.5	82.8	17.0	16.7	11.3‡	high	0.36	3	3	S	NR	4
477	oak, silky, northern	Q	NO	H	S6	SD7	J4	JD4	1050	500	3.1	3.5	64.0	72.0	14.7	8.8	4.7	low	0.31	4	—	S	NR	4
486	oak, tulip, brown	N, Q, G	BO	H	S2	SD2	J2	JD2	1150	900	5.6	7.3	—	—	—	—	9.0	—	0.44	4	—	S	NR	4
488	oak, tulip, red	Q	TO	H	S3	SD4	J2	JD2	1100	750	6.4	9.0	83.5	103.2	24.6	18.6	8.9	high	—	4	—	S	NR	4
505	penda, red	Q	PD	H	S2	SD2	J1	JD1	1150	1050	13.8	16.2	—	—	—	17.5	5.8	—	0.36	2	1	NS	R	2
512	peppermint, narrow-leaved	N, T, V	NL	H	S4	SD4	J3	JD2	1100	600	5.2	7.1	69.8	81.2	14.9	14.4	13.2‡	high	0.36	3	3	S	NR	4
533	pine, Carribbean	N, Q	PB	S	S6	SD6	J4	JD4	—	550	2.7	—	—	—	—	—	5.0	—	0.34	4	4	NS	R	4
		Fiji	PB	S	S6	SD6	J3	JD4	1020	600	—	—	—	—	—	—	3.9	—	—					
544	(pine) cypress, white	N, Q, S, V, W, Y	WC	S	S5	SD6	J3	JD3	850	700	4.6	6.1	43.2	37.4	12.9	7.3	2.6	low	0.30	2	1	NS	R	2

(*continued*)

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TABLE 1 (*continued*)

1	2	3	4	5	6		7		8		9		10		11		12	13	14	15		16	17	18
Index No.	Standard trade name	Origin	Species brand	Classification	Strength group		Joint group		Design density† (kg/m³)		Janka hardness, side grain (kN)		Tangential cleavage (N/mm)		Toughness (N·m)		Tangential % shrinkage	Tendency to split for unseasoned timber	% Unit tangential movement	Natural durability class§		Sapwood lyctus susceptible	Termite resistant	Marine borer durability
					G	D	G	D	G	D	G	D	G	D	G	D				In-ground contact	Outside above ground			
545	pine, hoop	N, Q, G	HP	S	S6	SD5	J4	JD4	800	550	2.7	3.4	30.6	54.2	13.6	4.5	3.8	low	0.23	4	4	NS	NR	4
551	pine, loblolly	N, Q	PL	S	S6	SD6	J3	JD3	1050	600	2.0	3.1	30.6	48.0	18.4	14.7	4.6	medium	0.31	4	4	NS	NR	4
554	pine, maritime	V, S, W	PM	S	S6	SD6	J4	JD4	1000	600	—	—	—	—	—	—	5.0	low	0.40	4	4	NS	R	4
559	pine, radiata	N, Q, S, T, V, W	PR	S	S6	SD6	J4	JD4	800	550	2.1	3.3	36.5	53.2	17.1	11.0	5.1	medium	0.27	4	4	NS	NR	4
		NZ			S7	SD6	J4	JD4	930	480	—	—	35.0	50.8	—	—	4.2	medium	—					
561	pine, slash	N, Q, W	PS	S	S5	SD5	J3	JD3	850	600	2.1	3.4	38.5	38.5	12.2	8.7	4.2	low	0.30	4	4	NS	R	4
615	satinash, Eungella, white	Q	ES	H	S4	SD5	J3	JD3	1050	650	—	—	—	—	—	—	—	—	—	3	—	S	—	—
618	satinash, grey	N, Q	GS	H	S5	SD6	J3	JD3	1100	650	4.8	5.6	75.4	79.3	24.3	16.2	5.7	low	0.28	3	—	S	NR	4
620	satinash, Kuranda	Q	KS	H	S3	SD3	J2	JD2	1100	750	5.9	9.1	101.5	89.8	27.3	20.4	6.8	low	0.35	3	—	S	—	—
635	satinay	Q	S	H	S3	SD3	J2	JD2	1100	850	5.2	8.3	74.2	99.8	15.1	10.2	10.0‡	high	0.35	2	1	NS	R	1
668	stringybark, Blackdown	Q	BN	H	S3	SD3	J1	JD1	—	1000	—	—	—	—	—	—	7.0	—	—	2	1	NS	R	—
671	stringybark, brown	N, S, W	BS	H	S3	SD3	J2	JD2	1100	850	5.4	7.5	71.6	89.1	16.2	18.8	10.4‡	high	0.33	3	2	NS	NR	4
676	stringybark, red	N, S, V	RS	H	S3	SD4	J2	JD2	1100	900	6.6	8.8	70.7	86.8	18.1	16.6	9.8‡	high	0.37	3	2	S	R	3
677	stringybark, silvertop	N, Q	SS	H	S2	SD2	J3	JD2	1050	850	5.5	8.8	—	—	—	—	8.0‡	—	—	3	3	NS	NR	4
680	stringybark, white	N, Q, V	WS	H	S3	SD3	J2	JD2	1100	850	6.4	9.0	74.4	96.6	19.7	21.7	10.6‡	high	0.36	2	2	NS	R	3
681	stringybark, yellow	N, V	YS	H	S3	SD3	J2	JD2	1150	900	6.6	8.6	70.0	73.5	19.4	18.7	7.5‡	medium	0.37	3	2	NS	R	3
688	tallowwood	N, Q	TW	H	S2	SD2	J1	JD2	1200	1000	7.1	8.6	73.8	67.2	19.7	19.7	6.1	low	0.37	1	1	S	R	3
723	turpentine	N, Q	TP	H	S3	SD3	J2	JD2	1050	950	6.6	11.6	70.3	107.8	18.5	14.7	13.0‡	high	0.35	2	1	NS	R	1
746	walnut, yellow	Q	YW	H	S4	SD5	J3	JD4	1050	600	3.2	4.8	65.0	70.0	8.3	9.6	3.8	low	0.27	4	—	S	NR	4
747	Wandoo	W	WG	H	S2	SD3	J1	JD1	1250	1100	9.9	14.7	79.8	65.8	24.0	19.4	3.9	low	0.29	1	1	NS	R	3

* Imported species (see Clause 4.2).

† Densities rounded to the nearest 50 (see Clause 4.9).

‡ Collapse susceptible species (see Clause 4.13).

§ Provisional classification (see Clause 4.16).

NOTE: The availability and source of particular species should be checked with local timber industry associations.

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4 EXPLANATION OF TABLE 1

4.1 General

This Clause gives an explanation of the column headings used in Table 1 together with a reference to the source document, as appropriate. Values of properties given in this Table are mean values for species group.

4.2 Index number (Column 1)

For Australian-grown timber species, the index number (without an asterisk) provides a cross-reference to AS/NZS 1148. This ensures that there is a correct linking between the standard trade name (Column 2) and the botanical name when required. Similarly for imported species (indicated by an asterisk), reference should be made to AS/NZS 1148 for the correct botanical name.

4.3 Standard trade name (Column 2)

The standard trade names listed are the preferred names. Their use is encouraged as a contribution to the more orderly identification, marketing and use of timber. Some alternative local names are given in AS/NZS 1148.

4.4 Origin (Column 3)

The State or States in which each Australian-grown timber occurs or country of origin is indicated by using the following code:

G—Papua New Guinea

N—New South Wales.

Nth Amer—North America

NZ—New Zealand

Q—Queensland

S—South Australia

T—Tasmania

V—Victoria

W—Western Australian

Y—Northern Territory

NOTE: (P) following an origin initial indicates plantation-grown timber.

4.5 Species brand (Column 4)

The species brand to be used when required follows the system given in AS 2209. For species not listed in that Standard, a circle has been drawn around the suggested abbreviation.

4.6 Classification (Column 5)

The letter 'H' indicates hardwood; or more specifically the timbers of the botanical group Angiosperms not all of which possess hard wood. The letter 'S' indicates the softwoods or timbers of the botanical group Gymnosperms, most of which are conifers.

4.7 Strength group (Column 6)

The strength group for each species in both the unseasoned or green (G) condition and the seasoned or dry (D) condition is given in accordance with the classification set out in AS 2878.

The relationship between strength group and visual grades is given in AS 2858.

Exceptions to the general rule are Douglas fir from North America in the dry condition and cypress pine. The appropriate stress grades for these species may be found by reference to AS 2858.

4.8 Joint group (Column 7)

For the purpose of joint design, all species have been classified into six groups, J1 to J6 for the unseasoned or green (G) condition, and six groups JD1 to JD6 for the seasoned or dry (D) condition, as shown in Table 2 and Table 3.

A1 NOTE: For information on mean density range for joint groups, see AS 1649.

TABLE 2

A1 **AVERAGE MINIMUM BASIC DENSITIES FOR JOINT STRENGTH GROUP IN UNSEASONED (GREEN) TIMBER**

Joint group	J1	J2	J3	J4	J5	J6
Basic density*, kg/m^3	750	600	480	380	300	240

* Basic density is the mass of an oven-dry specimen divided by its volume when green.

TABLE 3

A1 **AVERAGE MIMIMUM AIR-DRY DENSITIES FOR JOINT STRENGTH GROUPS IN SEASONED (DRY) TIMBER**

Joint group	JD1	JD2	JD3	JD4	JD5	JD6
Air-dry density at 12% m.c., kg/m^3	940	750	600	480	380	300

4.9 Design density (Column 8)

4.9.1 *Density of unseasoned timber*

The values tabulated for density of unseasoned or green (G) timber are design densities. For commercial unseasoned timber, these figures are approximate.

The unseasoned density figures are approximate being included only as a guide to enable self-weight of unseasoned members to be estimated in computing the dead load of a structure. Unseasoned density depends on the moisture content at the time of measurement. For the purposes of this tabulation it has been assumed that very little drying has taken place since the tree was felled and the densities given should be regarded as typical and not as exact values.

4.9.2 *Density of seasoned timber*

The values tabulated for the density of seasoned or dry (D) timber are design densities at 12% moisture content.

The density at 12% m.c. is the mass of a specimen divided by its volume when both measurements have been made when the specimen is at 12% m.c.

The values given apply before any reconditioning treatment. After reconditioning the values are 1% to 2% lower for most species but may be up to 10% lower in those that are collapse susceptible (see Column 12).

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Attention is drawn to the fact that rounded mean values are listed and cognizance should be given to the range of densities that could be encountered for any one species. The examples in Table 4 illustrate the point. Values given in Table 1 have been rounded up to the nearest 50 when the values of the last two digits are greater than or equal to 35 and less than or equal to 50, or greater than or equal to 85, and rounded down to the nearest 50 when the values of the last two digits are less than 35 or greater than 85. This has resulted in the values being generally in conformance with those given in AS 1720.1.

TABLE 4

EXAMPLES OF AIR-DRY DENSITY RANGES

Species	Air-dry density, kg/m^3	
	Mean	Range
ash, mountain	679	533 to 830
karri	905	814 to 1000
pine, radiata	535	420 to 650

4.10 Janka hardness, side grain (Column 9)

Janka hardness can refer to the difficulty in sawing, planing, resistance to wear or abrasion and the resistance to indentation. The figures given refer to this latter property, which generally is considered to be an indicator of the others.

The values are a measure of the force necessary to indent a timber specimen with a steel ball of 11.28 mm diameter to a depth of 5.64 mm (see CSIRO Technical Paper No 31). Data is the mean of the radial and tangential hardness values.

4.11 Tangential cleavage (Column 10)

Cleavage strength is a measure of a timber's tendency to split along the grain. This property influences its ability to take and hold fasteners under load. A related property is its tangential shrinkage (see AS 1720.1 and CSIRO Technical Paper No 31).

4.12 Toughness (Column 11)

Toughness is a measure of timber's ability to resist shocks and blows and is synonymous with impact strength (see CSIRO Technical Paper No 31). Data is the mean of the radial and tangential toughness values.

4.13 Tangential shrinkage (Column 12)

The percentage shrinkage values in drying from the unseasoned to 12% moisture condition are given for the tangential direction only. As a general rule, radial shrinkage is about half of the tangential value. It should also be noted that the values given apply before any reconditioning treatment (see CSIRO Technical Paper No 31).

Within this column a double dagger indicates that the particular species is normally reconditioned, this in turn suggests that it is susceptible to collapse, i.e. abnormally high shrinkage during its drying cycle.

NOTE: Further information on radial shrinkage is given in the references (a) and (b) of Note 1 to Clause 4.14.

4.14 Tendency for green wood to split (Column 13)

This column lists values of the parameter (α) defined by—

$$\alpha = \varepsilon^2/\gamma$$

where

ε = tangential shrinkage, in percent

γ = tangential cleavage strength of unseasoned timber, in newtons per millimetre, as measured by BS 373 or ASTM D143.

Species for which α is greater than 0.8 often have a high tendency to split, particularly in exposed locations; species for which α is less than 0.55 may be considered to have a negligible tendency to split.

NOTES:

1 Information on shrinkage and cleavage strength for specific species can be obtained from the following references:

(a) KINGSTON, R. S. T. and RISDON, C. J. E. *Shrinkage and Density of Australian and other South-West Pacific Woods*. Division of Forest Products Technological Paper, No 13, CSIRO, 1961.

(b) BUDGEN, B. *Shrinkage and Density of some Australian and South-East Asian Timbers*. Division of Building Research Technological Paper (2nd Series), No 38, CSIRO, 1981.

(c) BOLZA, E. and KLOOT, N. H. *The Mechanical Properties of 174 Australian Timbers*. Division of Forest Products Technological Paper, No 25, CSIRO, 1963.

2 It will be found that most Eucalypts and most hardwoods of dry sclerophyll forests that have a basic density of less than 700 kg/m^3 will have a splitting parameter (α) greater than 0.8; most softwoods and most rainforest hardwoods have a splitting parameter (α) less than 0.8.

4.15 Unit tangential movement (Column 14)

The percentage tangential movement for any change in moisture content between about 3% and within a few percent of the intersection point (typically between 20% and 25% moisture content) may be determined by multiplying the change in moisture content by the appropriate unit movement as listed.

4.16 Natural durability class (Column 15)

The classification system is taken from AS 5604, and is based on the average life expectancy (in years) of 50 × 50 mm test specimens, as given in Tables 5 for a species used in-ground or above-ground.

It is not intended that these ratings will be used as stand-alone criteria for determining the suitability of timber species for particular products, applications or environments. The performance and life expectancy of timber used in specific applications and environments will be greatly influenced by many other factors in addition to the natural durability ratings. These other factors include—

(a) presence or absence of preservative treatment;

(b) supplementary protection and maintenance;

(c) climate;

(d) environmental conditions;

(e) human influence;

(f) member size and detailing; and

(g) manufacturing process.

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As the structural capacity of a timber member or joint subjected to physical, chemical or biological hazards will diminish with time, allowance shall be made during the design to ensure the member or joint has adequate capacity at the end of the intended design life to ensure the intended structural function is achieved.

NOTE: Consideration should be given to the fact that the classification is very broad and it is not intended to distinguish between the relative merits of species in the same classification.

TABLE 5

NATURAL DURABILITY—PROBABLE LIFE EXPECTANCY*

Class	Probable in-ground life expectancy (years)	Probable above-ground life expectancy (years)
1	Greater than 25	Greater than 40
2	15 to 25	15 to 40
3	5 to 15	7 to 15
4	0 to 5	0 to 7

* The ratings in this Table are based on expert opinions and the performance of the following test specimens:

(a) In-ground: 50 × 50 mm test specimens at four sites around Australia.

(b) Above-ground: 35 × 35 mm test specimens at 11 sites around Australia.

NOTES:

1 As further reliable evidence becomes available, these ratings may require amending.

2 The heartwood of an individual piece of timber may vary from the species' nominated classification.

3 Above-ground conditions equate to outside above-ground subject to periodic moderate wetting when ventilation and drainage are adequate.

4 All untreated sapwood should be considered as natural durability class 4.

4.17 Sapwood lyctus susceptibility (Column 16)

The susceptibility of the sapwood of individual species to attack by the Lyctus borer is indicated in the following manner:

S—Susceptible

NS—Not susceptible

4.18 Termite resistant (Column 17)

Termite resistance of heartwood is classified as follows:

R—Resistant to termite.

NR—Not resistant to termite.

NOTE: Other species not listed or where there is no rating given should be assumed to be not resistant unless evidence to the contrary is provided by CSIRO or State Forestry departments.

4.19 Marine borer resistance ratings (Column 18)

The classification system is taken from AS 5604, and is based on the average life expectancy (in years), as given in Table 6, for a species used in southern marine waters.

TABLE 6

MARINE BORER RESISTANCE—PROBABLE LIFE EXPECTANCY

Class	Probable marine-borer-resistance life expectancy in southern waters (years)
1	Greater than 60
2	41 to 60
3	21 to 40
4	0 to 20, usually less than 5

NOTE: Marine borer resistance is based on natural round piles containing 350 mm diameter of heartwood in southern seas reaching from Perth in the west to Batemans Bay in the east. Only class 1 timbers and/or preservative-treated timber in accordance with AS 1604.1 can be expected to give reasonable service life in northern waters.

5 REFERENCES

The following references were consulted in the preparation of this Standard:

BEESLEY J., KEOUGH J.J., and MOULEN A.W., *Early burning properties of Australian building timber*, Div. Build. Res. Technical Paper (Second Series) No. 6, CSIRO Australia, 1974.

BOLZA E., and KLOOT N.H., *The mechanical properties of 174 Australian timbers*, Div. For. Prod. Technical Paper No. 25, CSIRO Australia, 1963.

BOOTLE Keith R., *Wood in Australia—Types, properties and uses, Second Edition*, Sydney: McGraw-Hill, 2005.

BUDGEN B., *Shrinkage and density of some Australian and South-East Asian timbers*, Div. of Build. Res. Technical Paper (Second Series) No. 38, CSIRO Australia, 1981.

DITCHBURNE N., KLOOT N.H., and RUMBALL B., *The mechanical properties of Australian-grown pinus radiata*, Div. Build. Res. Technical Paper (Second Series) No. 9, CSIRO Australia, D. Don. 1975.

JESSOME A.P., *Strength and related properties of woods grown in Canada*, Forestry Technical Report 21, Ottawa 1977.

MACK J.J., *Australian methods for mechanically testing small clear specimens of timber*, Div. Build. Res. Technical Paper (Second Series) No. 31, CSIRO Australia, 1979.

THORNTON J.D., JOHNSON G.C., and NGUYEN N.K., *Rivised natural durability ratings for the outer heartwood of nature Australian timbers in-ground contact*, 25th Forest Product Research Conference, Paper 1/18, 1996.

THORNTON J.D., JOHNSON G.C., and CREFFIELD J.W., *Hazard-based natural durability ratings and mean rates of bio-deterioration proposed for use in reliability-based durability design method*, 25th Forest Product Research Conference, Paper 1/18, 1996.

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APPENDIX A

F-GRADES FOR VISUALLY GRADED TIMBER

(Informative)

A1 F-GRADE

Three methods of assigning F-grade in Table A1 and A2 have been used, as follows:

(a) Method A: F-grade determined on the basis of small clear testing.

(b) Method B: F-grade determined on the basis of small clear testing and supported by in-grade testing.

(c) Method C: F-grade determined on the basis of in-grade testing in accordance with AS/NZS 4063.

A2 ASSIGNMENT OF F-GRADE

The historical F-grades in Tables A1 and A2 have been used successfully for many years for timber sourced from the old-growth forest. As more results of in-grade testing become available for a species, the stress grades in that Table for that species will be revised if necessary. Validation by in-grade testing in accordance with AS/NZS 4603 is recommended. In some instances, trade associations and research bodies hold validating information. Any stress grades evaluated through in-grade testing of full size structural material in accordance with AS/NZS 4063 should be taken to supersede all the above information.

NOTE: A more extensive list of timber species showing strength groups is given in Table 1.

TABLE A1

F-GRADES FOR VISUALLY GRADED TIMBER—SPECIES MIXTURES

Standard trade name	Moisture condition	Method of assigning F-grade (see Paragraph A1)	Strength group[1]	Stress grade (F-grade)				
				Structural grade number[2]				
				1	2	3	4	5
Softwood								
Hem-fir fir, amabilis fir, Californian red fir, grand fir, noble fir, white hemlock, western	Unseasoned	B	—	F7	F5	F4	—	—
	Seasoned	B	—	F11	F8	F7	F5	F4
Spruce-pine-fir (Canada) fir, alpine fir, balsam pine, jack pine, lodgepole pine, western white pine, western yellow spruce, black spruce, Engelmann spruce, red spruce, white	Unseasoned	B	—	F8	F7	F5	F4	—
	Seasoned	B	—	F8	F8	F7	F5	F4

(continued)

TABLE A1 (*continued*)

Standard trade name	Moisture condition	Method of assigning F-grade (see Paragraph A1)	Strength group[1]	Stress grade (F-grade) Structural grade number[2]				
				1	2	3	4	5
Hem-AmFir fir, amabilis hemlock, western	Unseasoned	—	—	—	—	—	—	—
	Seasoned	B	—	F11	F11	F8	F7	—
Hardwood								
balau *Shorea* spp.	Unseasoned	A	S3	F17	F14	F11	F8	—
	Seasoned	A	(SD3)	F27	F22	F17	F14	—
kapur *Dryobalanops* spp.	Unseasoned	A	S3	F17	F14	F11	F8	—
	Seasoned	A	SD4	F22	F17	F14	F11	—
keruing *Dipterocarpus* spp.	Unseasoned	A	S3	F17	F14	F11	F8	—
	Seasoned	A	SD3	F27	F22	F17	F14	—
kwila (merbau) *Intsia* spp.	Unseasoned	A	S2	F22	F17	F14	F11	—
	Seasoned	A	SD3	F27	F22	F17	F14	—
lumbayau (mengkulang) *Heritiera* spp.	Unseasoned	A	S5	F11	F8	F7	F5	—
	Seasoned	A	SD5	F17	F14	F11	F8	—
meranti, bakau *Shorea* spp.	Unseasoned	A	S4	F14	F11	F8	F7	—
	Seasoned	A	SD5	F17	F14	F11	F8	—
meranti, dark-red *Shorea* spp.	Unseasoned	A	S5	F11	F8	F7	F5	—
	Seasoned	A	SD6	F14	F11	F8	F7	—
meranti, light-red *Shorea* spp.	Unseasoned	A	S6	F8	F7	F5	F4	—
	Seasoned	A	SD7	F11	F8	F7	F5	—
meranti, white *Shorea* spp.	Unseasoned	A	S4	F14	F11	F8	F7	—
	Seasoned	A	SD5	F17	F14	F11	F8	—
meranti, yellow *Shorea* spp.	Unseasoned	A	S5	F11	F8	F7	F5	—
	Seasoned	A	(SD6)	F14	F11	F8	F7	—
mersawa *Anisoptera* spp.	Unseasoned	A	(S6)	F8	F7	F5	F4	—
	Seasoned	A	(SD6)	F14	F11	F8	F7	—
ramin *Gonystylus* spp.	Unseasoned	A	S4	F14	F11	F8	F7	—
	Seasoned	A	SD4	F22	F17	F14	F11	—
sepetir *Sindora* spp. *Copaifera* spp. *Pseudosindora* spp.	Unseasoned	A	S4	F14	F11	F8	F7	—
	Seasoned	A	(SD5)	F17	F14	F11	F8	—
taun *Pometia* spp.	Unseasoned	A	S4	F14	F11	F8	F7	—
	Seasoned	A	SD4	F22	F17	F14	F11	—

NOTES:

1 For classification into strength groups, see AS 2878.

2 For structural grade, see AS 2082 and AS 2858.

3 For timber poles, see AS 2209.

4 The strength groups given here are drawn from AS 2878 and those in brackets, e.g., '(SD4)', have provisional status as specified in AS 2878.

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TABLE A2

F-GRADES FOR VISUALLY GRADED TIMBER—SINGLE SPECIES

Index No.	Standard trade name	Moisture condition	Method of assigning F-grade (see Paragraph A1)	Strength group[1]	Stress grade (F-grade)					
					Structural grade number[2]					Pole timber[3]
					1	2	3	4	5	
8	Alder, rose	Unseasoned	A	S6	F8	F7	F5	F4	—	F11
		Seasoned	A	SD6	F14	F11	F8	F7	—	—
22	ash, alpine	Unseasoned	A	S4	F14	F11	F8	F7	—	F17
		Seasoned	B	SD4	F22	F17	F14	F11	—	—
30	ash, mountain	Unseasoned	A	S4	F14	F11	F8	F7	—	F17
		Seasoned	B	SD3	F27	F22	F17	F14	—	—
37	ash, silvertop	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	B	SD3	F27	F22	F17	F14	—	—
84	blackbutt	Unseasoned	B	S2	F22	F17	F14	F11	—	F27
		Seasoned	B	SD2	F34	F27	F22	F17	—	—
86	blackbutt, New England	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
87	blackbutt, W.A.	Unseasoned	A	S4	F17	F14	F11	F8	—	F22
		Seasoned	A	SD5	F14	F11	F8	F7	—	—
121	box, brush	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
127	box, grey, coast	Unseasoned	A	S1	F27	F22	F17	F14	—	F34
		Seasoned	A	SD1	F43	F34	F27	F22	—	—
165	brownbarrel	Unseasoned	A	S4	F14	F11	F8	F7	—	F17
		Seasoned	A	SD4	F22	F17	F14	F11	—	—
—	chengal	Unseasoned	A	S1	F27	F22	F17	F14	—	F34
		Seasoned	A	SD2	F34	F27	F22	F17	—	—
114	fir, Douglas, (North America)	Unseasoned	B	S5	F11	F8	F7	F5	F4	F14
		Seasoned	B	SD5	F11	F11	F8	F7	F5	—
	fir, Douglas (NZ)	Unseasoned	B	S6	F8	F7	F5	F4	—	F11
		Seasoned	B	SD6	F11	F11	F8	F7	F5	—
178	Candlebark	Unseasoned	A	S5	F11	F8	F7	F5	—	F14
		Seasoned	A	SD5	F17	F14	F11	F8	—	—
253	gum, blue, southern	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD2	F34	F27	F22	F17	—	—
254	gum, blue, Sydney	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
266	gum, grey	Unseasoned	A	S1	F27	F22	F17	F14	—	F34
		Seasoned	A	SD2	F34	F27	F22	F17	—	—
267	gum, grey, mountain	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD2	F34	F27	F22	F17	—	—

(continued)

TABLE A2 (*continued*)

Index No.	Standard trade name	Moisture condition	Method of assigning F-grade (see Paragraph A1)	Strength group[1]	Stress grade (F-grade)					
					Structural grade number[2]					Pole timber[3]
					1	2	3	4	5	
269	gum, manna	Unseasoned	A	S4	F14	F11	F8	F7	—	F17
		Seasoned	A	SD4	F22	F17	F14	F11	—	—
272	gum, mountain	Unseasoned	A	S4	F14	F11	F8	F7	—	F17
		Seasoned	A	SD5	F17	F14	F11	F8	—	—
279	gum, red, forest	Unseasoned	B	S3	F17	F14	F11	F8	—	F22
		Seasoned	B	SD4	F 17	F17	F14	F11	—	—
281	gum, red, river	Unseasoned	A	S5	F11	F8	F7	F5	—	F14
		Seasoned	A	SD5	F17	F14	F11	F8	—	—
284	gum, rose	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD4	F22	F17	F14	F11	—	—
289	gum, shining	Unseasoned	A	S3	F14	F11	F8	F7	—	F17
		Seasoned	A	SD4	F22	F17	F14	F11	—	—
293	gum, spotted	Unseasoned	B	S2	F22	F17	F14	F11	—	F27
		Seasoned	B	SD2	F34	F27	F22	F17	—	—
310	hardwood, Johnstone River	Unseasoned	A	S2	F22	F17	F14	F11	—	F27
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
139	hemlock, western	Unseasoned	B	S6	F8	F7	F5	F4	—	F11
		Seasoned	B	SD6	F11	F11	F8	F7	F5	—
322	ironbark, grey	Unseasoned	B	S1	F27	F22	F17	F14	—	F34
		Seasoned	B	SD1	F34	F34	F27	F22	—	—
325	ironbark, red	Unseasoned	B	S2	F22	F17	F14	F11	—	F27
		Seasoned	B	SD3	F27	F22	F17	F14	—	—
326	ironbark, red, broad-leaved	Unseasoned	B	S1	F27	F22	F17	F14	—	F34
		Seasoned	B	SD1	F43	F34	F27	F22	—	—
327	ironbark, red, narrow-leaved	Unseasoned	B	S2	F22	F17	F14	F11	—	F27
		Seasoned	B	SD3	F27	F22	F17	F14	—	—
341	jarrah	Unseasoned	A	S4	F14	F11	F8	F7	—	F17
		Seasoned	A	SD4	F22	F17	F14	F11	—	—
344	karri	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD2	F34	F27	F22	F17	—	—
—	kempas	Unseasoned	A	S2	F22	F17	F14	F11	—	F27
		Seasoned	A	SD2	F34	F27	F22	F17	—	—
384	mahogany, red	Unseasoned	A	(S2)	F22	F17	F14	F11	—	F27
		Seasoned	A	(SD3)	F27	F22	F17	F14	—	—
432	marri	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
434	messmate	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	B	SD3	F27	F22	F17	F14	—	—

(*continued*)

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TABLE A2 (*continued*)

Index No.	Standard trade name	Moisture condition	Method of assigning F-grade (see Paragraph A1)	Strength group[1]	Stress grade (F-grade)					
					Structural grade number[2]					Pole timber[3]
					1	2	3	4	5	
477	oak, silky, northern	Unseasoned	A	S6	F8	F7	F5	F4	—	F11
		Seasoned	A	SD7	F11	F8	F7	F5	F4	—
486	oak, tulip brown	Unseasoned	A	S2	F22	F17	F14	F11	—	F27
		Seasoned	A	SD2	F34	F27	F22	F17	—	—
488	oak, tulip, red	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD4	F22	F17	F14	F11	—	—
505	penda, red	Unseasoned	A	S2	F22	F17	F14	F11	—	F27
		Seasoned	A	SD2	F34	F27	F22	F17	—	—
512	peppermint, narrow-leaved	Unseasoned	A	S4	F14	F11	F8	F7	—	F17
		Seasoned	A	SD4	F22	F17	F14	F11	—	—
533	pine, Carribbean	Unseasoned	A	S6	F8	F7	F5	F4	—	F11
		Seasoned	B	SD6	F11	F11	F8	F7	F5	—
544	(pine) cypress, white	Unseasoned	C	S5	—	—	F7	F5	F4	F14
		Seasoned	C	SD6	—	—	F7	F5	F4	—
545	pine, hoop	Unseasoned	A	S6	—	—	F8	F7	F5	F11
		Seasoned	C	SD5	—	—	F8	F7	F5	—
551	pine, loblolly	Unseasoned	A	S6	F8	F7	F5	F4	—	F11
		Seasoned	B	SD6	F11	F11	F8	F7	F5	—
554	pine, maritime	Unseasoned	A	S6	F8	F7	F5	F4	—	F11
		Seasoned	B	SD6	F11	F11	F8	F7	F5	—
559	pine, radiata (Australia)	Unseasoned	A	S6	F8	F7	F5	F4	—	F11
		Seasoned	B	SD6	F11	F11	F8	F7	F5	—
	pine, radiata (New Zealand)	Unseasoned	A	S7	F8	F7	F5	F4	—	F8
		Seasoned	B	SD6	F11	F11	F8	F7	F5	—
561	pine, slash	Unseasoned	A	S5	F11	F8	F7	F5	F4	F14
		Seasoned	B	SD5	F14	F14	F11	F8	F7	—
615	satinash, Eungella, white	Unseasoned	A	S4	F14	F11	F8	F7	—	F17
		Seasoned	A	SD5	F17	F14	F11	F8	—	—
618	satinash, grey	Unseasoned	A	S5	F11	F8	F7	F5	—	F14
		Seasoned	A	SD6	F14	F11	F8	F7	—	—
620	satinash, Kuranda	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
635	satinay	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
668	stringybark, Blackdown	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
671	stringybark brown	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—

(*continued*)

TABLE A2 (*continued*)

Index No.	Standard trade name	Moisture condition	Method of assigning F-grade (see Paragraph A1)	Strength group[1]	Stress grade (F-grade)					
					Structural grade number[2]					Pole timber[3]
					1	2	3	4	5	
676	stringybark, red	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD4	F22	F17	F14	F11	—	—
677	stringybark, silvertop	Unseasoned	A	S2	F22	F17	F14	F11	—	F27
		Seasoned	A	SD2	F34	F27	F22	F17	—	—
680	stringybark, white	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
681	stringybark yellow	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
688	tallowwood	Unseasoned	A	S2	F22	F17	F14	F11	—	F27
		Seasoned	A	SD2	F34	F27	F22	F17	—	—
723	turpentine	Unseasoned	A	S3	F17	F14	F11	F8	—	F22
		Seasoned	A	SD3	F27	F22	F17	F14	—	—
746	walnut, yellow	Unseasoned	A	S4	F14	F11	F8	F7	—	F17
		Seasoned	A	SD5	F17	F14	F11	F8	—	—
747	wandoo	Unseasoned	A	S2	F22	F17	F14	F11	—	F27
		Seasoned	A	SD3	F27	F22	F17	F14	—	—

NOTES:

1 For classification into strength groups, see AS 2878.

2 For structural grade, see AS 2082 and AS 2858.

3 For timber poles, see AS 2209.

4 The strength groups given here are drawn from AS 2878 and those in brackets, e.g., '(SD4)', have provisional status as specified in AS 2878.

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AMENDMENT CONTROL SHEET

AS 1720.2—2006

Amendment No. 1 (2006)

CORRECTION

SUMMARY: This Amendment applies to Clauses 2 and 4.8, and Tables 2 and 3.

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